

G. Narayanamma Institute of Technology & Science (for Women)
(Autonomous)

Shaikpet, Hyderabad- 500 104

IV-B.Tech I-Semester Supplementary Examinations, April-May-2024

DATA SCIENCE USING R
(Common to CSM & CSD)

Max. Marks: 70

Time: 03 Hours

Note:

1. Question paper comprises of **Part A** and **Part B**.
2. **Part A** is compulsory which carries 10 marks. Answer all questions in Part A.
3. **Part B** (for 60 marks) consists of **five questions** with **“either” “or”** pattern. Each question carries 12 marks and may have a,b,c as sub questions. The student has to answer any one full question.

PART-A

(Answer 05 questions. Each question carries 2 marks)

Q.No.	Question	Marks	CO	B L
Q.1	a) Describe how to load data in R.	[02]	CO1	[L1]
	b) Write a short note on Invalid values and Outliers.	[02]	CO2	[L2]
	c) Describe the methods used for validating the models.	[02]	CO3	[L2]
	d) How does hierarchical clustering work?	[02]	CO4	[L4]
	e) What is time series similarity search?	[02]	CO5	[L1]

END OF PART A

PART-B

(Answer 05 full questions. Each question carries 12 marks)

Q.No.	Question	Marks	CO	B L
Q.2(a)	Write a program in R to read the data from an XML file.	[06]	CO1	[L1]
(b)	Discuss the stages of a Data Science project in detail with a neat sketch.	[06]	CO1	[L1]

OR

Q.3(a)	Explain Data types in R.	[06]	CO1	[L2]
(b)	Explain some essential R commands and functions used for data exploration and summary statistics?	[06]	CO1	[L1]

Q.4(a)	Write about a data frame? Explain various operations on data frame.	[06]	CO2	[L3]
(b)	Explain about the concept of Finding the Missing Values.	[06]	CO2	[L3]

OR

Q.5(a)	Relate the purpose of descriptive statistics in exploratory data analysis.	[06]	CO2	[L4]
(b)	Differentiate between data cleaning, data validation and explain data cleaning techniques.	[06]	CO2	[L2]

- Q.6(a)** Describe building models using a single variable and explain why we add more variables to a model. [06] CO3 [L3]
- (b)** What is KDD (Knowledge Discovery in Databases), and what is its significance in data mining and machine learning? [06] CO3 [L2]

OR

- Q.7(a)** Mention some common evaluation metrics used to assess the performance of machine learning models? [06] CO3 [L2]
- (b)** Describe the main objective of KDD cup 2009 conference. [06] CO3 [L1]

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- Q.8(a)** Compare and contrast between simple linear regression, multiple linear regression. [06] CO4 [L4]
- (b)** Explain K-means clustering algorithm, with an example. [06] CO4 [L3]

OR

- Q.9(a)** What is the primary purpose of logistic regression, and how does it differ from linear regression? [06] CO4 [L2]
- (b)** How do you choose an appropriate distance metric for a given clustering problem? [06] CO4 [L1]

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- Q.10(a)** Can you outline some common applications of time series analysis in real-world scenarios. [06] CO6 [L2]
- (b)** List the common file formats for storing time series data, and explain how to read them in to R. [06] CO5 [L1]

OR

- Q.11(a)** Explain the importance of understanding the residual component in time series analysis. [06] CO6 [L2]
- (b)** Explain forecasts using Exponential smoothing. [06] CO5 [L1]

END OF PART B
END OF THE QUESTION PAPER